Presentation script: results and interpretation

Slide 1

Hi everyone. As we've been discussing, consumer sentiment can look much weaker than headline economic indicators suggest. The question is whether economic news helps explain that gap, and whether it adds useful information when predicting sentiment. The dataset here combines national news, local reporting, and forum-submitted stories. That gives a way to examine how different types of coverage relate to sentiment. The main finding is that news tone helps a simple economic model, but that benefit weakens once broader conditions and previous sentiment are included.

Slide 2

This chart explains the starting point. The dark line shows actual Michigan sentiment, and the green line shows the prediction using unemployment, inflation, and GDP growth. Since January 2022, actual sentiment averages 14.85 index points below that prediction. This shows that the baseline is incomplete. Household pressures, expectations, and other missing influences could contribute, so the gap cannot be attributed to media alone. This chart uses a separate, longer evaluation covering 81 months.

Slide 3

The analysis uses 670 economic titles. NPR provides national reporting, Colorado Sun provides state-level coverage, and Hacker News contributes forum-submitted stories. This source mix brings together different parts of the economic conversation. Each title receives an automated label for adverse, mixed, or favorable economic conditions. Labels are averaged within each source and then combined with equal source weights to create a monthly tone index. There are 47 balanced media months, but only 45 match finalized sentiment outcomes under the timing rules. The first 24 form the initial training sample, leaving 21 forecast evaluations.

Slide 4

The classifier comparison highlights a measurement problem. Agreement with automated references is 68 percent during development, but falls to 53.3 percent on a new set of 300 titles. These are different evaluation sets, so the numbers need that qualification. Still, the result warns against assuming that development performance will carry over to new economic headlines. The primary index uses a separate set of saved automated labels. Its conclusions remain tentative because errors in those labels can affect the monthly index and, in turn, the regression results.

Slide 5

The source comparison helps explain what the tone measure represents. Housing words appear in about 31 percent of Colorado Sun titles, compared with 10 percent of Hacker News titles. Inflation-rate wording shows the opposite pattern: about 20 percent for Hacker News and 3 percent for Colorado Sun. So is that a difference in bias, or in the conditions being covered? A source covering housing hardship may sound negative because the underlying conditions are difficult. Tone therefore needs to be interpreted alongside topic selection. Equal source weights reduce the influence of collection volume, but they cannot remove differences in the issues each source covers.

Slide 6

The regression results show why the comparison matters. The simple model uses unemployment, annual PCE inflation, and GDP growth. Adding news tone reduces forecast error from 12.07 to 10.82. That is roughly a 10 percent observed improvement, but its uncertainty interval includes zero. The broader model includes accumulated prices, real-income growth, and historical-era controls. Without tone, its error is 5.82. With tone, error rises slightly to 6.06. Because both use the same controls and evaluation months, this comparison isolates the additional forecasting contribution of tone. The stronger overall performance is already present without it. That substantially weakens the initial interpretation that news provides a distinct predictive signal.

Slide 7

Previous sentiment makes the test more demanding. Simply predicting last month unchanged gives an error of 4.09. Broader controls plus previous sentiment give 3.98. Adding tone gives 4.32. Recent sentiment therefore provides a strong benchmark that the original news measure does not improve on. One possible explanation is overlapping information: headlines may describe pressures already captured by the economic inputs or previous sentiment. Another is that the tone measure is noisy. The small sample cannot distinguish these explanations confidently. What it does show is that the additional forecasting benefit is not dependable in these comparisons.

Slide 8

The factual example explains why tone should not automatically be interpreted as bias. In the July 2020 GDP release, a 32.9 percent annualized decline corresponds to about a 9.5 percent decline over the actual quarter. Both numbers describe the same change, but they can create different impressions without the units. The relevant question is whether the headline supplies enough context to interpret the number correctly. A tone score captures whether the economic direction sounds favorable or adverse. Establishing misleading framing requires a factual benchmark. These selected cases illustrate that distinction, but do not measure how readers respond.

Slide 9

Taken together, the results highlight the importance of household conditions, sentiment persistence, and the way news is measured. The source comparison shows that different outlets emphasize different economic pressures. News tone therefore becomes more meaningful when it is interpreted alongside topic selection and factual context. For forecasting, the key lesson is to assess what news adds beyond economic conditions and previous sentiment. Its apparent contribution changes once those stronger comparisons are introduced. The practical value is a framework that combines economic data with headline analysis to monitor changes in the economic conversation and investigate sentiment gaps. That provides a useful foundation for improving the measures and testing future forecasts. Thank you.